

Component-weighted bias in rotation curve model selection:

A proxy-chain audit of χ^2 -based ΔAIC statistics

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Abstract. Rotation curve fitting via global χ^2 minimisation followed by Akaike Information Criterion (ΔAIC) comparison is the standard tool for gravitational model selection across the SPARC galaxy sample. We show that this pipeline contains a structural measurement bias: in bulge-dominated galaxies, χ^2 weight concentrates on baryonic inner regions rather than the dark-matter or modified-gravity sector under test. We formalise this through a proxy-chain audit $\Phi \rightarrow V(r) \rightarrow \chi^2 \rightarrow \Delta\text{AIC}$, introduce a component-weighted bias metric B_{bulge} , and compute it for all 175 SPARC galaxies. We find a statistically significant Spearman correlation $\rho = 0.176$ ($p = 0.020$) between B_{bulge} and ΔAIC , rising to $\rho = 0.474$ ($p = 0.013$) in the LATENT (high- L) regime. Among galaxies with $B_{\text{bulge}} < 0.1$, the EFC model wins or ties in 91.9% of cases; for $B_{\text{bulge}} > 0.2$, this drops to 44.4%. The bias is model-agnostic and affects NFW, MOND/RAR, and EFC frameworks alike. We propose an RCMB-compliant protocol reporting $\Delta\text{AIC}_{\text{inner}}$ and $\Delta\text{AIC}_{\text{outer}}$ separately.

Keywords: rotation curves; model selection; χ^2 weighting; AIC bias; SPARC; proxy-chain audit; RCMB

Introduction

Rotation curve decomposition has been the central observational test for dark matter and alternative gravity theories since Rubin & Ford (1970). The standard workflow fits model-predicted circular velocities to $\{r_i, V_{\text{obs},i}, \sigma_i\}$ data via least-squares minimisation, producing a reduced χ_r^2 and, in comparative analyses, a ΔAIC score between competing models.

The inner region of spiral galaxies is observationally well-resolved and statistically advantageous: small radii yield many data points with small photon-noise uncertainties. Oman et al. (2015) and Li et al. (2020) document that this same inner region is the locus of the rotation-curve “diversity problem”, where predicted and observed profiles disagree systematically. Sofue (2017) recommends dividing fitting radii by component to avoid degeneracies. The ANN-based decomposition of Pedrosa et al. (2026) likewise identifies bright-bulge galaxies as problematic for parametric fitting.

However, none of these works identify the consequence at the level of *model selection metrics*: if the inner region dominates χ^2 , and the inner region is baryonically dominated, then ΔAIC does not report model performance in the regime where dark matter or modified gravity governs the dynamics. It reports baryonic fitting quality. This paper formalises that measurement bias, computes it for the full SPARC 175 sample, and proposes a corrected test protocol.

The proxy chain and where information is lost

Formal proxy chain

Model evaluation via rotation curves passes through four transformation steps:

$$\Phi_{\text{model}}(r) \xrightarrow{(1)} V_{\text{model}}(r) \xrightarrow{(2)} \chi^2 \xrightarrow{(3)} \Delta\text{AIC} \quad (1)$$

Each step compresses information:

1. **Component mixing.** The observable is total circular velocity:

$$V_{\text{total}}^2(r) = V_{\text{gas}}^2 + V_{\text{disk}}^2 + V_{\text{bulge}}^2 + V_{\text{model}}^2(r) \quad (2)$$

The model term V_{model} contributes fractionally, and its fractional contribution vanishes in bulge-dominated regions.

2. **Radial weighting.** The standard χ^2 statistic

$$\chi^2 = \sum_{i=1}^N \frac{(V_{\text{obs},i} - V_{\text{model},i})^2}{\sigma_i^2} \quad (3)$$

weights each data point by $w_i = \sigma_i^{-2}$. In high-surface-brightness (HSB) spirals, inner data points have high V_{obs} and low σ , yielding systematically large w_i relative to outer points.

3. **Single-number compression.** ΔAIC collapses the full residual profile to one scalar, suppressing information about *where* the model succeeds or fails along the curve.

Consequence

In a galaxy with a prominent bulge (Hubble type Sa–Sb), $V_{\text{bulge}}^2 \gg V_{\text{model}}^2$ for $r \lesssim 3 \text{ kpc}$. A model with higher parametric freedom in that region will absorb the bulge residuals and produce a lower χ^2 , *regardless of its physical validity at $r \gtrsim 5 \text{ kpc}$* where halo physics is

dominant. NFW benefits from a free concentration parameter c that can adjust the inner profile; EFC-R uses a fixed screening parameter $k = 0.415$ calibrated on the full SPARC sample with LSB weight. The resulting ΔAIC therefore conflates baryonic fitting flexibility with model validity.

Formal bias metrics

Inner-weight fraction B

For each galaxy with N data points at radii $\{r_i\}$ and weights $\{w_i = \sigma_i^{-2}\}$, define a critical radius r_c separating the baryonic-inner from the halo-outer regime (default: $r_c = 3 \text{ kpc}$):

$$B = \frac{\sum_{r_i < r_c} w_i}{\sum_{i=1}^N w_i} \quad (4)$$

$B \rightarrow 1$ indicates that virtually all χ^2 weight is carried by the inner region; $B \rightarrow 0$ indicates an outer-dominated fit. A galaxy with $B > 0.5$ is *inner-biased*: its global ΔAIC primarily reflects inner-region fit quality.

Component-weighted bias B_{bulge}

A sharper metric accounts for the fractional bulge contribution at each point:

$$B_{\text{bulge}} = \frac{\sum_{i=1}^N w_i \cdot \frac{V_{\text{bulge},i}^2}{V_{\text{total},i}^2}}{\sum_{i=1}^N w_i} \quad (5)$$

B_{bulge} measures the weighted average fraction of χ^2 that is “spent” on bulge-dominated data points. If the test hypothesis is that two models differ in their *halo* description, B_{bulge} quantifies how much of the reported ΔAIC is irrelevant to that hypothesis.

Bias condition

We define a fit as *model-selection biased* if

$$B_{\text{bulge}} > 0.4 \quad \text{and} \quad \Delta\text{AIC}_{\text{outer}} \text{ has opposite sign to } \Delta\text{AIC}_{\text{global}} \quad (6)$$

This condition is testable from the raw SPARC data without any new observations.

Results: SPARC 175 bias census

We compute B and B_{bulge} for all 175 galaxies in the SPARC catalogue (Lelli et al. 2016) using the decomposed rotation curve data (V_{gas} , V_{disk} , V_{bulge} from *Spitzer* $3.6 \mu\text{m}$ photometry), cross-referenced with ΔAIC values from Kill-Test v6 (Magnusson 2026b).

Bulge prevalence

Of 175 galaxies, 143 (81.7%) have $B_{\text{bulge}} = 0$ (no bulge component in the SPARC decomposition). The remaining 32 galaxies carry non-zero bulge weight, of which 18 have $B_{\text{bulge}} > 0.2$ and 11 have $B_{\text{bulge}} > 0.4$.

The population-level statistics are:

- B : median = 0.265, mean = 0.326
- B_{bulge} : median = 0.000, mean = 0.058

The low median for B_{bulge} reflects the SPARC sample’s composition: predominantly late-type spirals and dwarfs with negligible bulges. The bias is concentrated in the minority of early-type spirals where it matters most for model selection.

Correlation with ΔAIC

Over the full sample, the Spearman rank correlation between B_{bulge} and ΔAIC is

$$\rho(B_{\text{bulge}}, \Delta\text{AIC}) = 0.176, \quad p = 0.020 \quad (7)$$

Positive ΔAIC favours $\Lambda\text{CDM}/\text{NFW}$; negative favours EFC. The correlation confirms that higher bulge contamination shifts ΔAIC toward NFW preference.

Crucially, the purely radial metric shows *no* correlation: $\rho(B, \Delta\text{AIC}) = 0.002$ ($p = 0.98$). It is specifically the *component-weighted* bias—not mere inner-region weighting—that predicts model selection outcomes.

Per-regime breakdown

Table 1 shows that the bias manifests differently across EFC entropy-flow regimes (Magnusson 2026a):

Table 1: Bias metrics and $B_{\text{bulge}}\text{--}\Delta\text{AIC}$ correlation by EFC regime. The LATENT regime shows the strongest bias effect.

Regime	N	$\langle B \rangle$	$\langle B_{\text{bulge}} \rangle$	$\langle \Delta\text{AIC} \rangle$	ρ	p
FLOW	62	0.309	0.035	−201.5	−0.382	0.002
TRANSITION	86	0.411	0.032	−22.9	−0.122	0.26
LATENT	27	0.093	0.193	+11 166	+0.474	0.013

In the LATENT regime—where EFC predicts the model is outside its domain of validity— B_{bulge} correlates strongly with ΔAIC ($\rho = 0.474$, $p = 0.013$). This is precisely where bulge-dominated early-type spirals concentrate, and where NFW’s concentration-parameter freedom absorbs bulge residuals most effectively.

The FLOW regime shows a *negative* correlation ($\rho = -0.382$, $p = 0.002$): among FLOW galaxies the few with prominent bulges actually favour EFC more. This is consistent with EFC’s entropy-gradient mechanism operating in the inner region where the baryonic potential gradient is steepest.

Model selection by bulge bias

The most direct test of the bias hypothesis is the EFC win rate stratified by B_{bulge} :

Table 2: EFC win/tie rate stratified by bulge bias.

B_{bulge} range	N	EFC win or tie
< 0.1 (negligible bulge)	149	137 (91.9%)
> 0.2 (significant bulge)	18	8 (44.4%)

The drop from 91.9% to 44.4% is striking. In the low-bulge population, ΔAIC overwhelmingly favours EFC; in the high-bulge population, the results are inconclusive. This is exactly the signature expected from bulge-weight contamination of a halo-sector test.

Extreme cases

Table 3 shows the five most and least bulge-biased galaxies in the sample:

Table 3: Top 5 most and least bulge-biased galaxies.

Galaxy	B	B_{bulge}	ΔAIC	Regime	Winner
<i>Most bulge-biased:</i>					
UGC 02916	0.20	0.838	−1180	TRANSITION	EFC
NGC 4217	0.11	0.810	−91	FLOW	EFC
NGC 7814	0.02	0.654	−4	TRANSITION	TIE
UGC 06614	0.00	0.634	+3	TRANSITION	TIE
UGC 05253	0.29	0.617	+4713	LATENT	LCDM
<i>Least bulge-biased:</i>					
CamB	1.00	0.000	+2	TRANSITION	TIE
DDO 154	0.49	0.000	−87	FLOW	EFC
DDO 161	0.07	0.000	−20	FLOW	EFC
DDO 168	0.88	0.000	−13	FLOW	EFC
DDO 170	0.02	0.000	−14	FLOW	EFC

Note that UGC 02916 ($B_{\text{bulge}} = 0.84$) and NGC 4217 ($B_{\text{bulge}} = 0.81$) still show EFC wins despite extreme bulge bias. This suggests that even when bulge contamination is severe, EFC’s entropy-gradient mechanism can track the baryonic potential. Conversely, UGC 05253 ($B_{\text{bulge}} = 0.62$, LATENT) shows $\Delta\text{AIC} = +4713$ favouring LCDM—a case where the combination of high bulge bias and LATENT-regime physics produces extreme NFW advantage.

RCMB-compliant test protocol

The Robust Causal Measurement Basis (RCMB) framework, introduced for CMB–void stacking in Magnusson (2026c, DOI: [10.6084/m9.figshare.31942677](https://doi.org/10.6084/m9.figshare.31942677)), requires that the proxy chain from physical target to extracted statistic be audited for (i) epicenter fidelity, (ii) compression loss, and (iii) component contamination. Applied to rotation curve model selection:

1. **Epicenter audit.** The intended epicenter is the halo/modified-gravity sector at $r \gtrsim 5$ kpc. In 25.1% of SPARC galaxies $B > 0.5$, indicating the actual measurement epicenter has drifted to the inner region.
2. **Decompression requirement.** A single global ΔAIC is insufficient. Report $\Delta\text{AIC}_{\text{inner}}$ ($r < r_c$) and $\Delta\text{AIC}_{\text{outer}}$ ($r \geq r_c$) separately.

3. **Component isolation.** Outer-zone fits should use $V_{\text{residual}}^2 = V_{\text{total}}^2 - V_{\text{baryon}}^2$ where the baryonic contributions are fixed from photometry, testing only the dark-matter or modified-gravity sector directly.

Discussion

This is a measurement-level finding, not a model preference

The bias described here is model-agnostic. NFW benefits from concentration-parameter freedom in the inner zone; MOND/RAR benefits from the baryonic Tully–Fisher calibration being driven by the same inner data. EFC-R is penalised because its screening parameter is fixed. None of these outcomes reflects halo-sector performance. Correcting for the bias does not *favour* any particular model; it removes a confound that currently distorts comparisons.

Relation to prior work

Oman et al. (2015) documented inner-region diversity as a physical puzzle. Li et al. (2020) showed systematic inner mismatches in RAR fits. Sofue (2017) recommended component-separated fitting for practical reasons. Pedrosa et al. (2026) demonstrated that ANN-based decomposition recovers different mass budgets in bright-bulge galaxies.

None of these works connect the inner-region problem to a formal measurement-bias at the level of the ΔAIC model-selection statistic. The present paper makes that connection explicit, provides quantitative metrics (B , B_{bulge}), demonstrates a statistically significant correlation with model selection outcomes across 175 galaxies, and proposes a corrected protocol.

The B vs B_{bulge} distinction

A key result is that the purely radial metric B shows zero correlation with ΔAIC ($\rho = 0.002$), while the component-weighted B_{bulge} shows a significant one ($\rho = 0.176$, $p = 0.020$). This means the bias is not about where the data points are—it is about what physical component dominates the signal at those points. A galaxy can have most of its χ^2 weight at $r < 3$ kpc (high B) yet show no bias if the inner region is gas- or disk-dominated rather than bulge-dominated.

Limitations

The ΔAIC values used here are global (not split by radius). A full test of the bias condition (Eq. 6) requires re-fitting each model to inner and outer subsets separately, which is straightforward but computationally intensive. The current analysis demonstrates that the *precondition* for bias (B_{bulge} correlating with ΔAIC) is satisfied; the full inner/outer decomposition is a natural follow-up.

Reproducibility

All analysis code is deposited with this preprint: `compute_bias_metrics.py` (Python 3, numpy, scipy). Input data: SPARC rotation curves (Lelli et al. 2016, <http://astroweb.cwru.edu/SPARC/>) and Kill-Test v6 ΔAIC values (Magnusson 2026b, DOI: [10.6084/m9.figshare.31964847](https://doi.org/10.6084/m9.figshare.31964847)). Per-galaxy derived metrics (B , B_{bulge} , regime, ΔAIC) are deposited as `bias_metrics_sparc175.json`.

Frozen parameters: $r_c = 3$ kpc; $\sigma_{\text{min}} = 1.0 \text{ km s}^{-1}$; $k = 0.415$ (EFC-R screening, frozen per Kill-Test v6 governance).

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